DPUSNDXEgo

DPUSNDXEgo simulates engine and wind and roll noises of the car by using sound data from Allersmeier.

1. Inputs:

RPM	engine speed (rounds per minute)
AcceleratorPedal	position of the accelerator pedal [0 .. 1]
v_kmh	speed of the vehicle (kilometers per hour)
Horn	horn (0/1)
Indicator	indicator relais sound (0/1)
Starter	ignition sound (0/1)
Rumpeln	rumble sounds [0 .. 4]

2. Parameters:

AudioData	file with sound data
Motor[X,Y,Z]	position of motor sound source
Wind[X,Y,Z]	position of wind sound source
Misc[X,Y,Z]	position of others (horn, indicator, rumble) sounds source
WetRoad	additional noise for wet roads (0/1)
GlobalVolume	global volume of the sounds [0..1] (default: 1)
WindVolume	volume of wind sound [0 .. 1] (default: 1)
LeerBoost	boost factor of loudness of idle engine (default: 1)
LastBoost	boost factor of loudness of load engine (default: 1)
MiscBoost	boost factor of loudness of misc sounds (default: 1)
AA	„AntiAliasing“ (0/1) Smoothens steps in engine speed and position of the accelerator pedal when activated. (default: 0)
DoMixRPM	mix multiple engine speed sounds? (default: 0)
BufferMS	length of the sample buffer in ms (default: 16)
BufferN	number of buffers for the sound source (default: 10)
WindSoundOn	play wind sound samples? (0/1, default: 1)
WindVMaxSample	maximum wind speed (default: 179.9)
StartSampleLeer	first sample to use for idle engine (default: 1)
StartSampleLast	first sample to use for load engine (default: 1)
LeerSampleOffset	offset for samples (do not change) (default: 2)
LastSampleOffset	offset for samples (do not change) (default: 2)

3. Outputs: (for debugging)

CurrentLastSampleRPM	load engine speed of sound sample playing
CurrentLeerSampleRPM	idle engine speed of sound sample playing
CurrentWindSampleV	wind speed of sound sample playing

4. Dependencies:

This DPU needs sound simulation data with premade samples. Only *BMW520E34V17.SIM* is supported at the moment.

5. Functional principle:

DPUSNDXEgo loads sound samples for engine (idle and load), wind and other vehicle sounds of the file and mixes them according to the inputs and parameters before playing them over OpenAL. Setting the input of Starter to 1 while the engine speed is below 500 rpm the ignition sound is played back. A rumble sound can be played the same way by setting the *Rumpeln* input to a value unequal to zero. Playback is done as long the input is set to that value. Likewise, setting the input of *Horn* to 1 results in a horn noise, setting the input of Indicator to 1 results in a indicator relay sound.

Changing the factor of *LastBoost* or *LeerBoost* can be used to increase or decrease loudness of the engine sound.

If *AA* is set to 1 the input of *RPM* and *AcceleratorPedal* is smoothened. This may result in a difference between the displayed values in the instrument cluster and the playback sounds but may improve the transition between different sound samples.

6. Example:

```
# vehicle position
DPUSNDXListener SNDXListener
{
      Computer = {SNDX};
      Index = 20;
};

Connections =
{
      VDyn.v -> SNDXListener.v,
      VDyn.yaw -> SNDXListener.yaw,
      VDyn.pitch -> SNDXListener.pitch,
      VDyn.roll -> SNDXListener.roll,
      VDyn.X -> SNDXListener.X,
      VDyn.Y -> SNDXListener.Y,
      SCNX.ZOut -> SNDXListener.Z
};

# vehicle sound
```

```
DPUSNDXEgo SNDXEgoSoundSim
{
	Computer = {SNDX};
	Index = 200;
};

Connections =
{
	VDyn.v_kmh -> SNDXEgoSoundSim.v_kmh,
	VDyn.RPM -> SNDXEgoSoundSim.RPM,
	Mockup.AcceleratorPedal -> SNDXEgoSoundSim.AcceleratorPedal
};

# “mix” left and right indicator signal
DPUMinMax ~Indicator
{
	Computer = {SNDX};
	Index = 200;
	Min = 0;
};
Connections =
{
	Mockup.IndicatorLeft -> ~Indicator.In1,
	Mockup.IndicatorRight -> ~Indicator.In2,
	~Indicator.Out -> SNDXEgoSoundSim.Indicator,

	Mockup.Horn -> SNDXEgoSoundSim.Horn
	Mockup.Starter -> SNDXEgoSoundSim.Starter
};
```